

Easy AR Geo Spatial Anchors

Easy AR Geo Spatial Anchors Documentation

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1. Introduction

Easy AR Geo Spatial Anchors is an innovative geospatial app designed to revolutionize how users interact with the physical world through augmented reality (AR). This powerful application allows users to create, manage, and experience AR content tied to specific real-world locations. With the ability to anchor digital objects in precise geographical coordinates, Easy AR Geo Spatial Anchors bridges the gap between virtual and physical spaces, offering enhanced navigation, immersive experiences, and contextually relevant information.

The app is ideal for a wide range of applications, from urban exploration and tourism to educational tools, real estate, and even retail. By leveraging GPS data and advanced AR technology, it ensures that the digital assets remain consistently anchored in place, regardless of time or environmental changes. Whether you're guiding users through a museum with interactive exhibits or adding AR directions for an outdoor event, Easy AR Geo Spatial Anchors makes it simple and intuitive to create engaging and context-aware AR experiences. Built for flexibility and ease of use, Easy AR Geo Spatial Anchors caters to both developers and non-technical users, making AR more accessible than ever before.

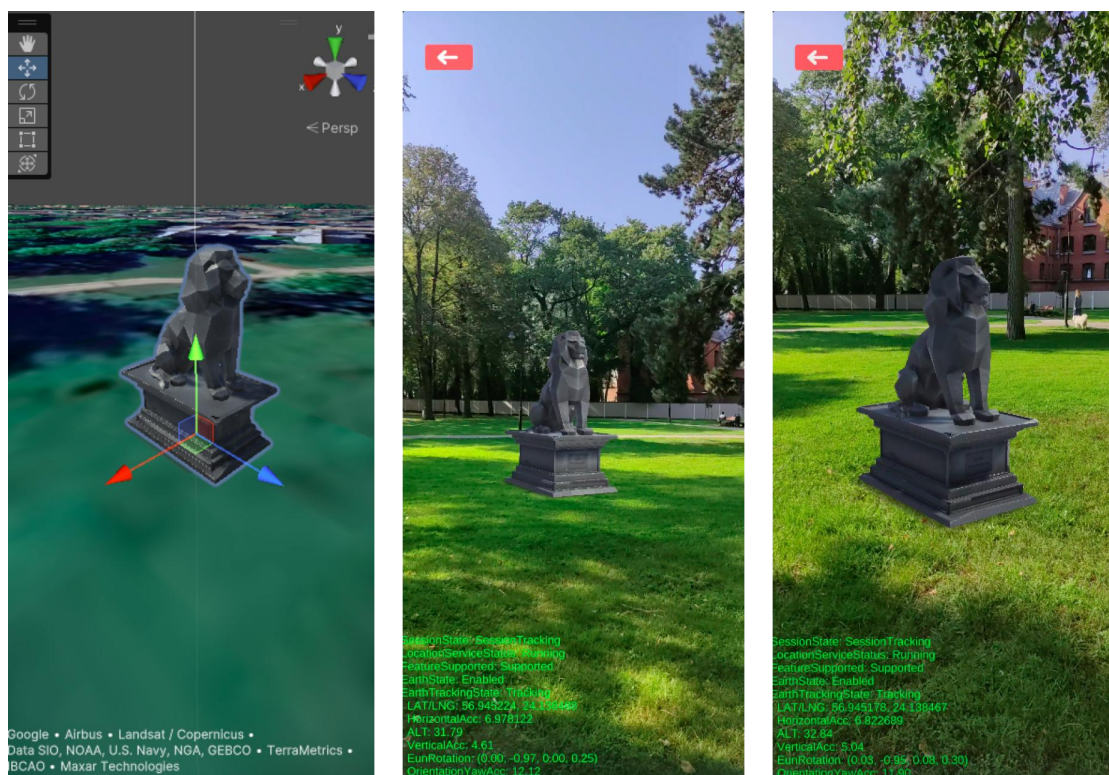
2. Features in Easy AR

- 1) Place AR content precise location given geo coordinates
- 2) Editor and Runtime AR Object Placement
- 3) Automatic Lights & Reflections
- 4) No Coding Skills Needed
- 5) Compatible with iOS and Android
- 6) Free Demo Experience (Balloon Hunt)

3. Demos

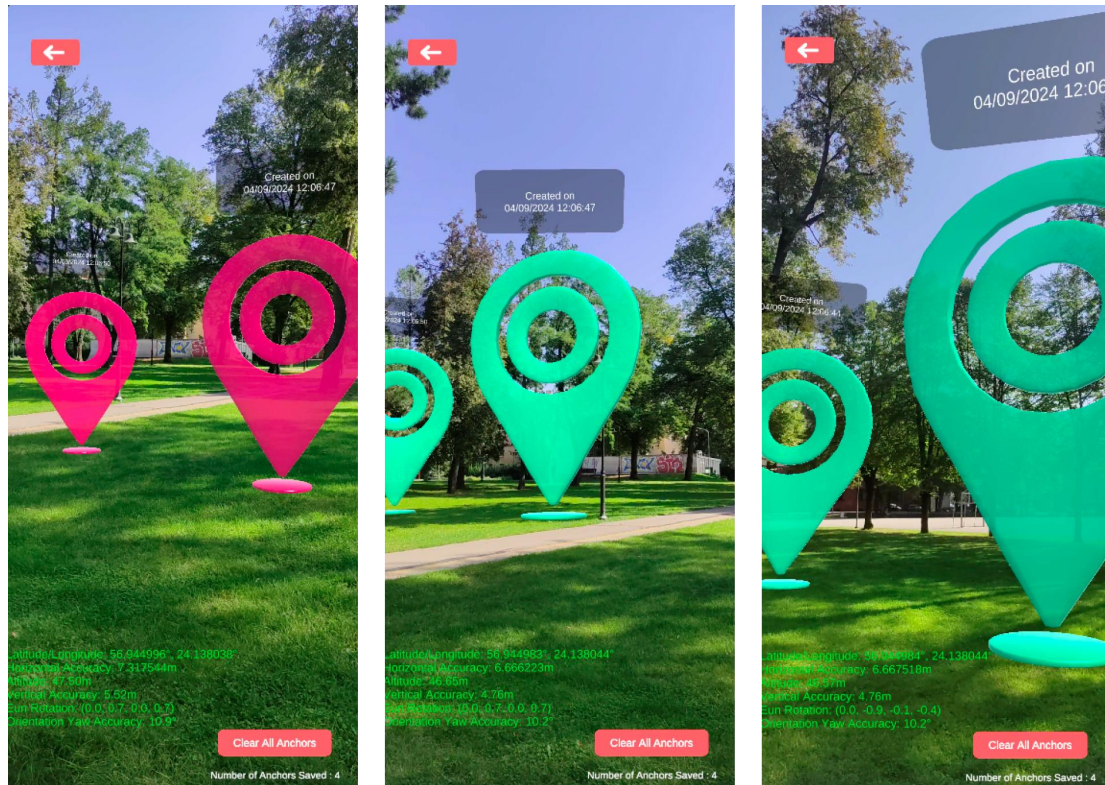
Easy AR Geo Spatial Anchors provides 3 different demos which easy to use, place 3D objects any where in the world using Geo Spatial API with simple build. These demos can configure as your requiremet and enjoy the app. The demos that are used as follows,

3.1. Place Anchor from Editor



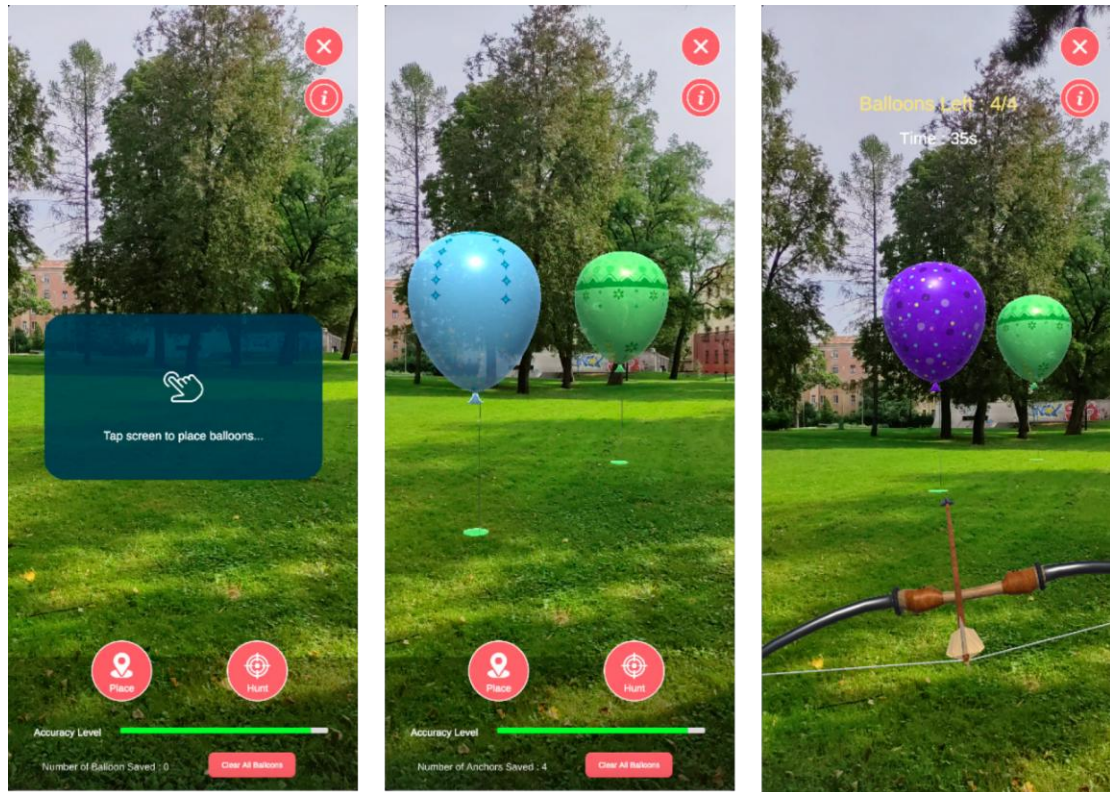
Place Anchor from Editor demo provide tile map of earth inside Unity editor. You can give preferred coordinates using latitude, longitude and altitude to place and experience AR content tied to specific real-world locations. With this ability to anchor digital objects in precise geographical coordinates.

3.2. Place Anchors Runtime



From this demo you can place geo spatial anchor runtime simply by tap on the mobile screen. This demo provide place and retriive anchor runtime, though no need to use unity editor to place anchor, But still you can change default AR content by editor. Maximum number of anchor place limits to 20 and anchors saved mobile itself automatically after placing. Anchor created date and time show in simple card top of AR content. Saved anchors automatically retrived and popup by opening the demo scene.

3.3. Balloon Hunt Demo



Balloon Hunt demo made using geo spatial anchor runtime demo gemified version, which you can place and hunt balloon after game started within given time. After place balloons the locations saved in mobile device. So you are able to start hunting any time in saved location. You will have bow and arrows to hunt balloons.

4. Package details and Dependancies

4.1. Unity Version

- Unity 2021.3. or higher LTS versions

4.2. AR Foundation version

- AR Foundation package 4.2.10

4.3. ARCore XR Plugin version

- ARCore XR Plugin 4.2.10

4.4. ARKit XR Plugin version

- ARKit XR Plugin 4.2.10

4.5. ARCore Extensions

- ARCore Extensions 1.37.0

4.6 Cesium for Unity

- Cesium Version 1.6.4

5. Setting up Easy AR Geo Spatial Anchors Package in Unity

5.1. Download Packages

Download following package

- Cesium 1.12.0 (<https://github.com/CesiumGS/cesium-unity/releases/download/v1.12.0/com.cesium.unity-1.12.0.tgz>)

5.2 Setting up Unity Project

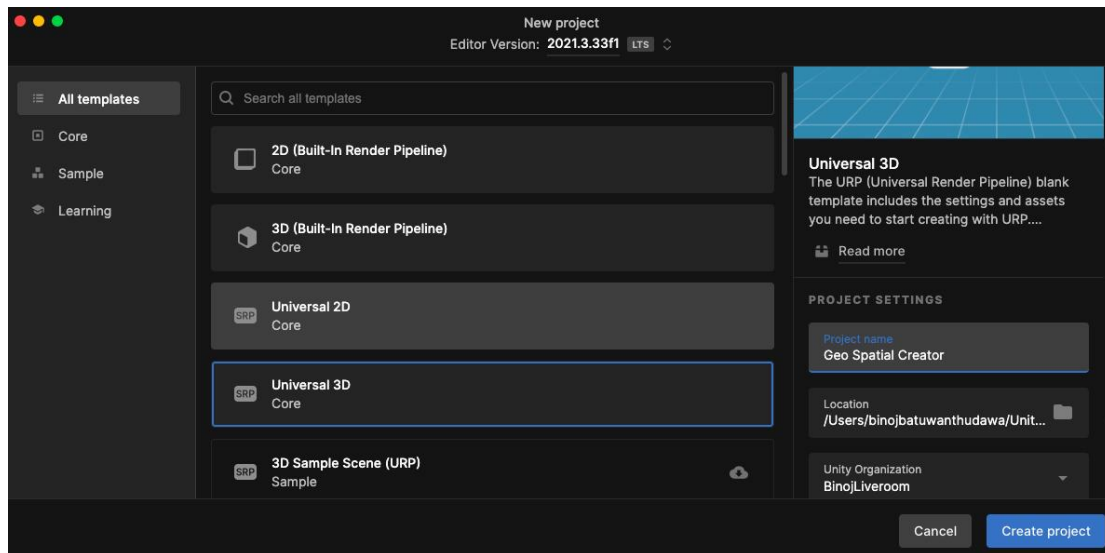
We highly recommend using **Unity (LTS)** verified versions. Initially we built the package using the following Unity version and used the **AR Foundation package**, **ARCore XR Plugin** and **ARKit XR Plugin**.

5.2.1 Unity Version

- **Unity** 2021.3. or higher LTS versions

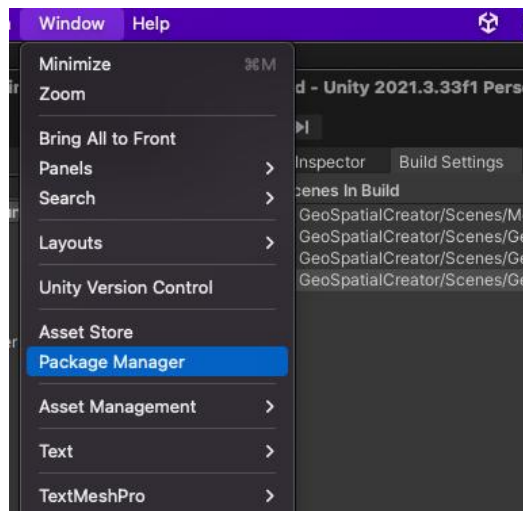
In the Unity hub, under the Installs tab you can add this verified version.

- Open Unity hub and create new project usign 3D URP template.
- Give project name as you desire. I will name this project as **Geo Spatial Creator**.
- Then hit create project button.



5.2.2 Installing Dependancies Package Manager

- Go to window->Package Manger



- Select Unity Registry and click AR
- Import following packages

- **AR Foundation version**

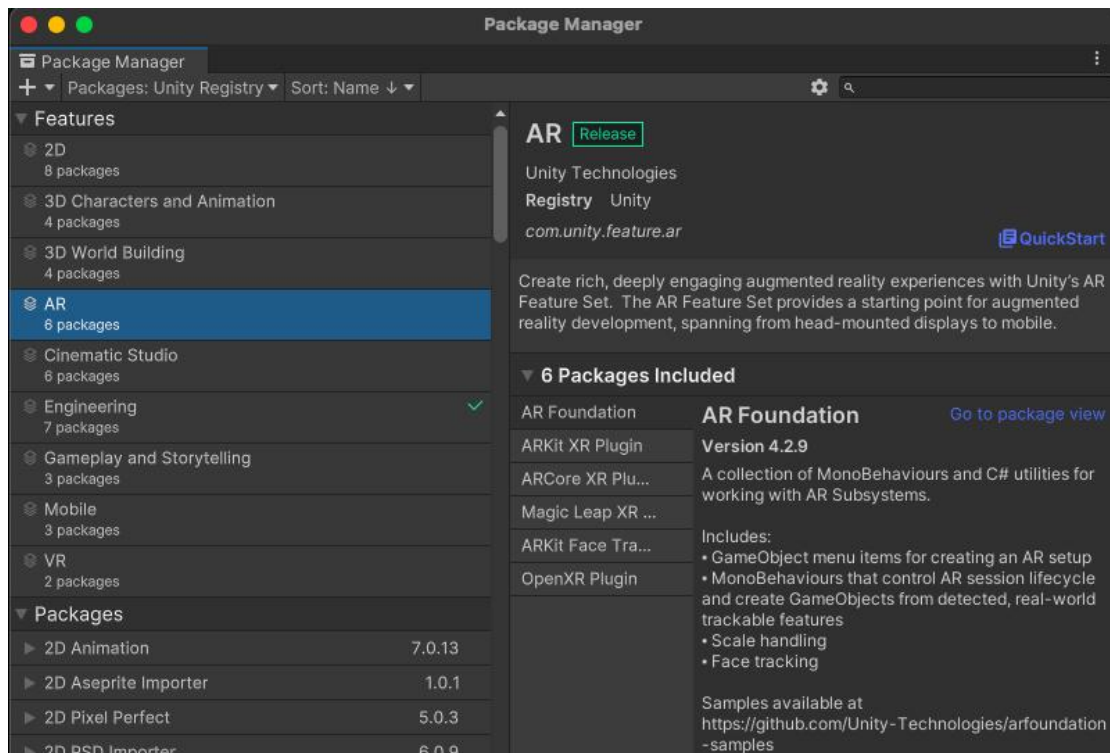
- AR Foundation package 4.2.10

- **ARCore XR Plugin version**

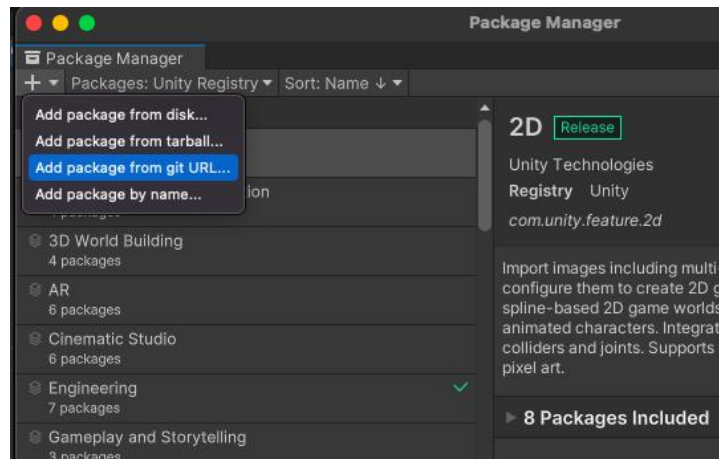
- ARCore XR Plugin 4.2.10

- **ARKit XR Plugin version**

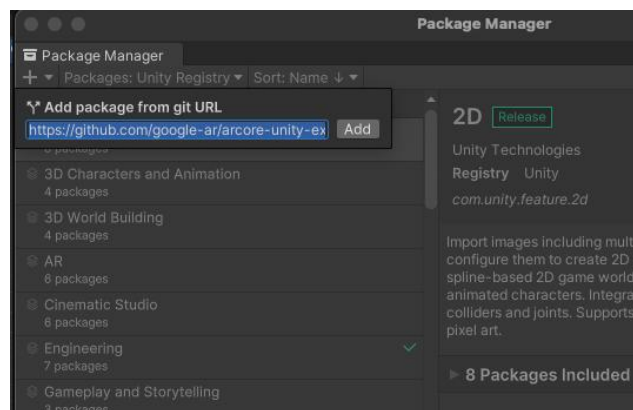
- ARKit XR Plugin 4.2.10



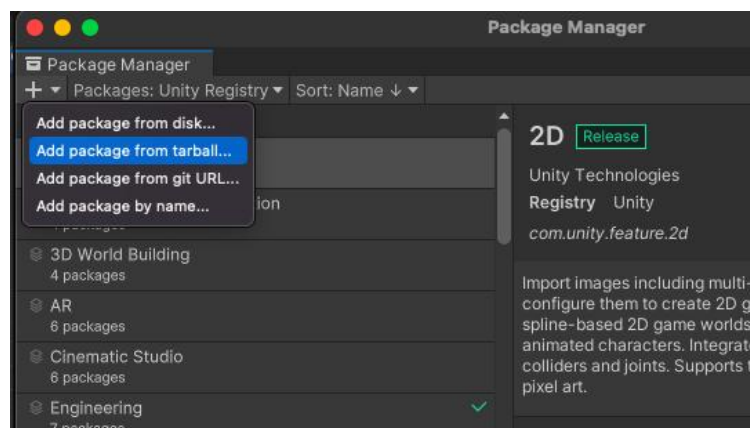
- Click the  button and choose the Add package from git URL... option from the drop-down menu



- Copy and paste following URL into text field under Add package from git URL and press Add button.



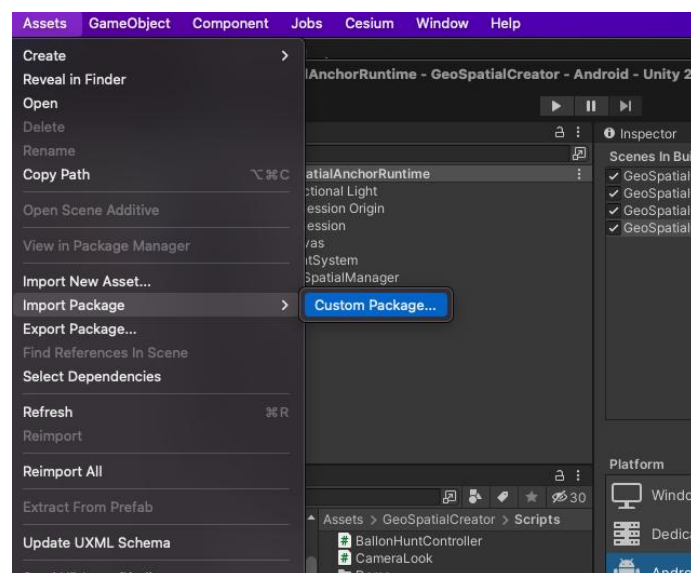
- Then AR Extension package will be added to your project.
- Click plus(+) mark again and select add packages from tarball.



- Then select cesium package that downloaded before and click open.
- Next import Textmeshpro resources from Unity Packages
- Click ARCore Extensions. Under Samples click Geospatial Sample and import sample.

5.3 Add Geo Spatial Creator package into Unity package.

- Go to Assets -> Import Package -> Custom Package...
- Select package and click open.



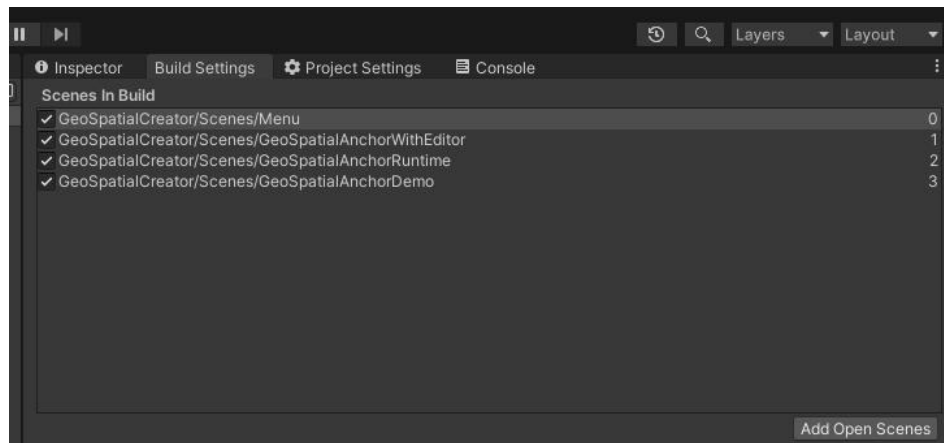
5.4. Configure Project

5.4.1. Go to File->Build settings

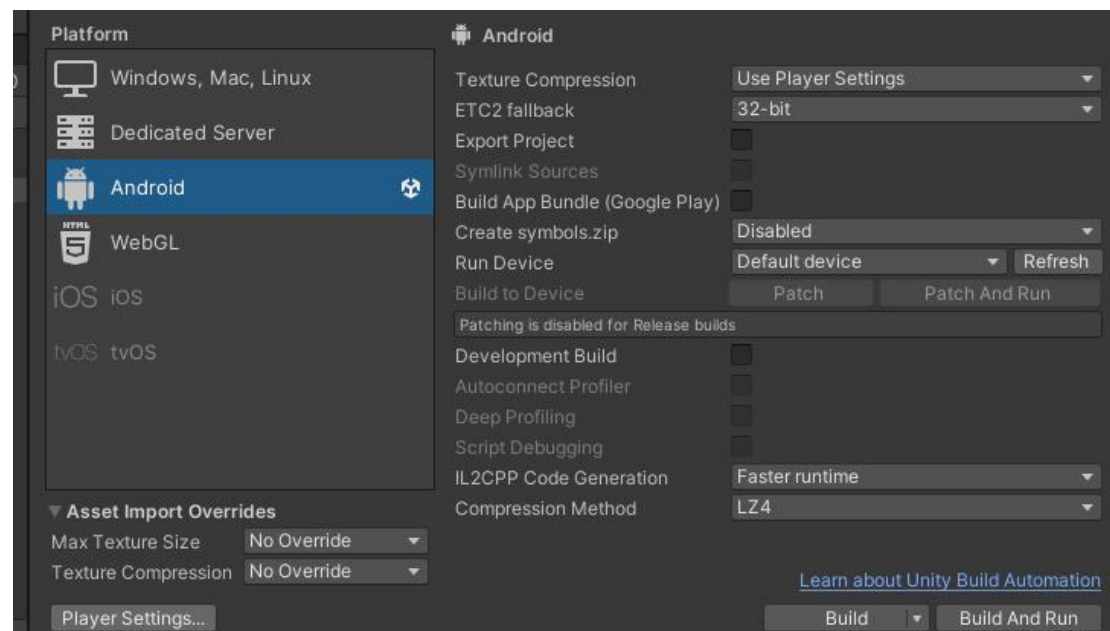
- Go to **GeoSpatialCreator->Scenes** folder and drag and drop the five scenes to build settings.

Build index order should be like below

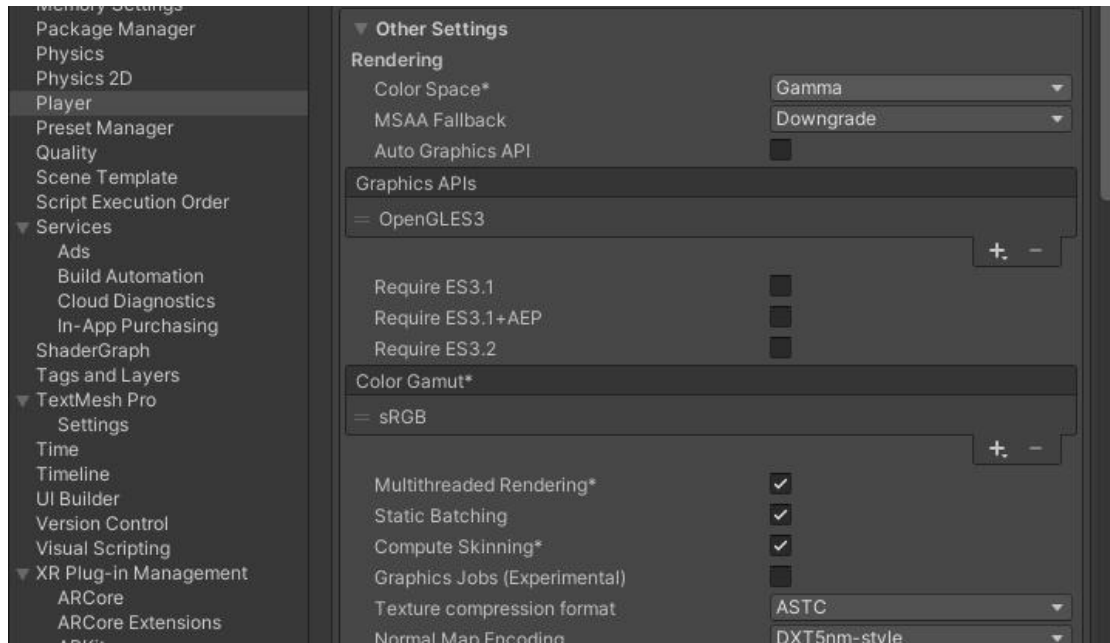
- 0 – GeoSpatialCreator/Scenes/Menu
- 1 – GeoSpatialCreator/Scenes/GeoSpatialAnchorDemo
- 2 – GeoSpatialCreator/Scenes/GeoSpatialAnchorRuntime
- 3 – GeoSpatialCreator/Scenes/GeoSpatialAnchorWithEditor



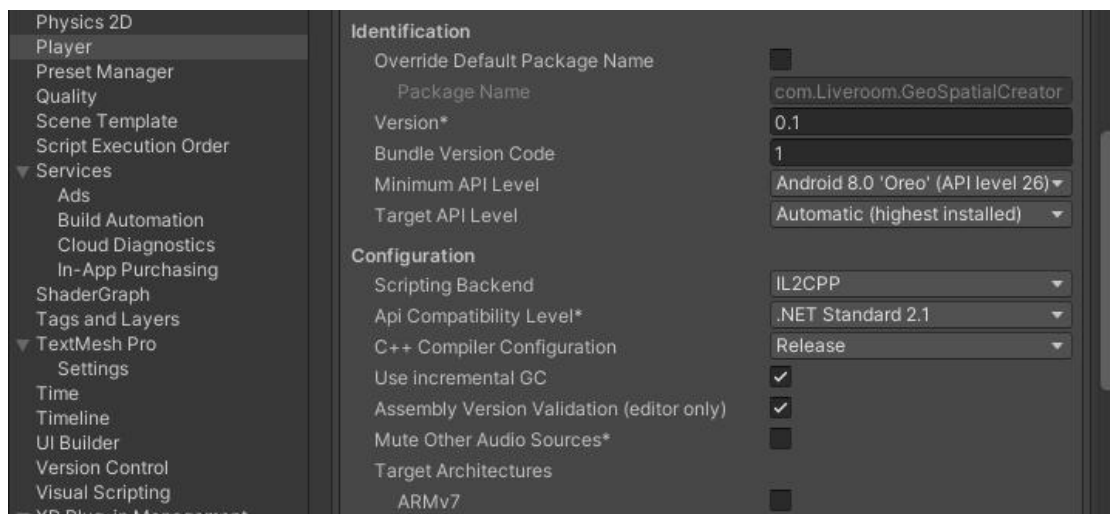
- Switch target Platform you desire to build either android or IOS.



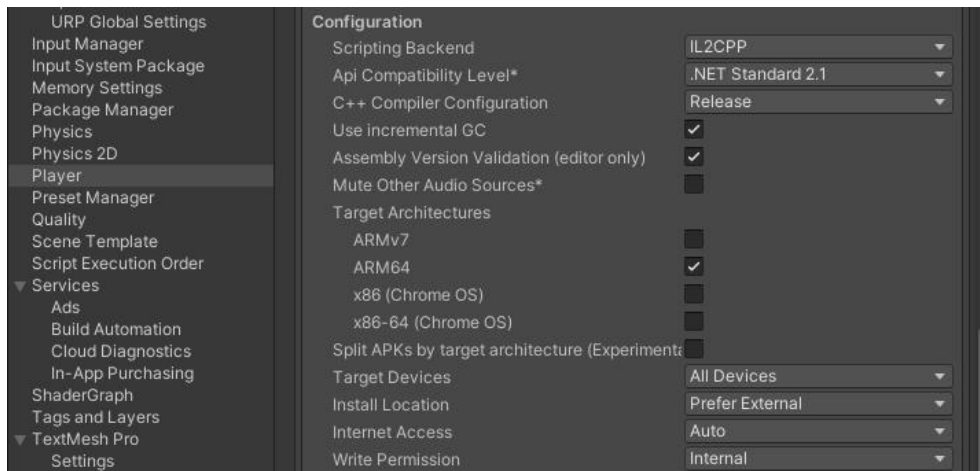
- Then Go to **Player setting** -> **Player** -> **Other settings**
- Under **Rendering** change **Color Space** to **Gamma**
- Disable **Auto Graphics API**
- Click **Vulkan Graphic API** and remove



- Then under **Identification** upgrade the **minimum API level** to **API level 26**.
- Then under **Configuration** change **scripting backend** to **IL2CPP**



- For Target Architectures disable ARMv7 and **enable ARM64**.



- If your **desired platform** is **iOS**, switch the target platform to iOS. Go to **Player setting** → **XR Plug-in Management** & go to **IOS tab**. Under plug in providers enable ARKit.
- Then Go to **Player setting** → **Player** → **Other settings** in the IOS tab. Check under **Identification** below context are filled
bundle identifier → com.companyName.productName
- Camera usage description → Use AR Camera
- Location Usage description → Use Location
- If your desired platform is **android** go to **Edit > ProjectSettings > Player > Other Settings** then set the android **Minimum API** level to **API level 26** and **remove Vulkan Graphic API**
- Here we go...now you can **build** the app by pressing the build **button** in the build setting. But before that you have to setup Geo Spatial API in Google Cloud Project. Let's focus on this next chapter.

3. Setting up Geo Spatial API in Google Cloud Project

3.1. Configure Google Cloud Project

- Redirect to google cloud console (<https://console.cloud.google.com/>)
- Click dropdown next to Google Cloud label



You're working in [GeoSpatialNew](#)

- It appears project window and click new project at the top right corner.



- It will appear New Project window. Give Project name as you desire and click create button.

Google Cloud

Search (/) for resources, docs, products, and more

New Project

You have 14 projects remaining in your quota. Request an increase or delete projects. [Learn more](#)

[MANAGE QUOTAS](#)

Project name *

GeoSpatialCreator

?

Project ID: geospatialcreator-435208. It cannot be changed later. [EDIT](#)

Location *

No organization

BROWSE

Parent organization or folder

CREATE

CANCEL

- Then it will appear your newly created project home page. In Quick access you can see API and services. Click and redirect to it.

Create a VM

Run a query in BigQuery

Create a GKE cluster

Create a storage bucket

Try Gemini →

Quick access

API APIs & Services

IAM & Admin

Billing

Compute Engine

Cloud Storage

BigQuery

VPC network

Kubernetes Engine

- Click enable API and Services

API APIs & Services

APIs & Services

+ ENABLE APIS AND SERVICES

Enabled APIs & services

Library

Credentials

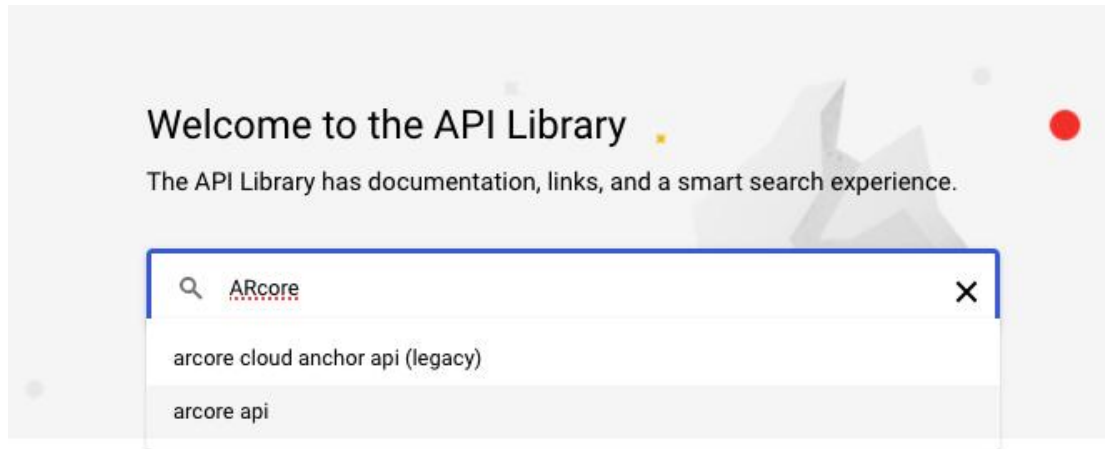
OAuth consent screen

Page usage agreements

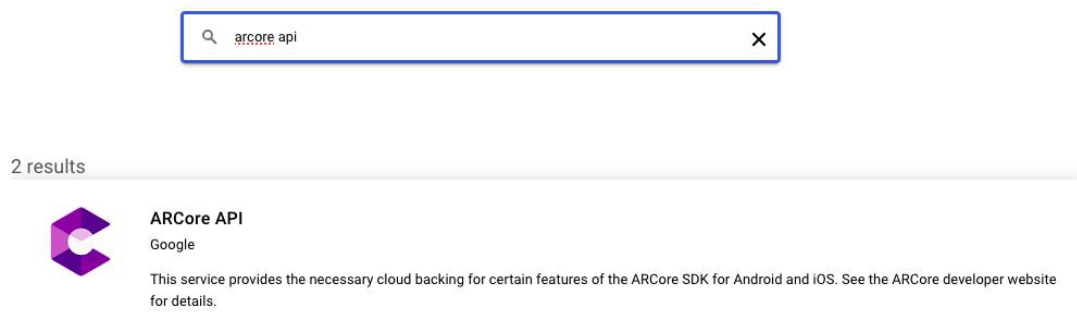
Traffic

No data is available for the selected time frame.

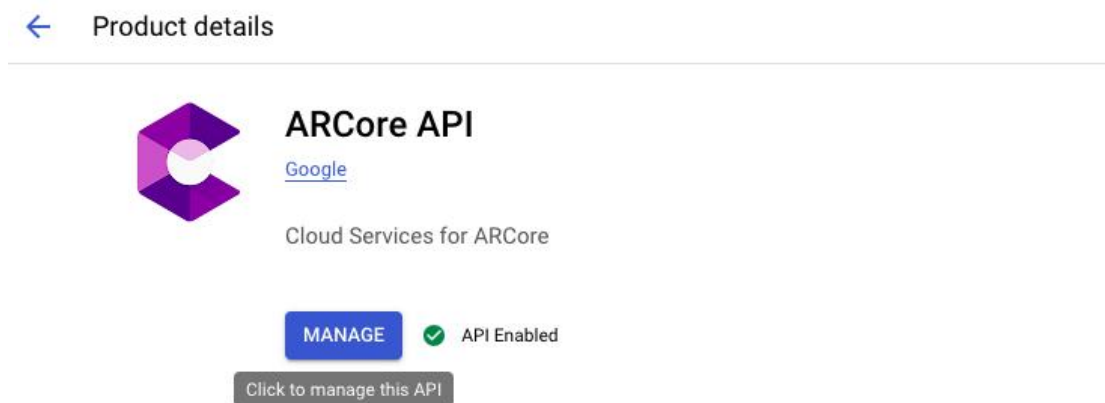
- In search bar find ARCore API.



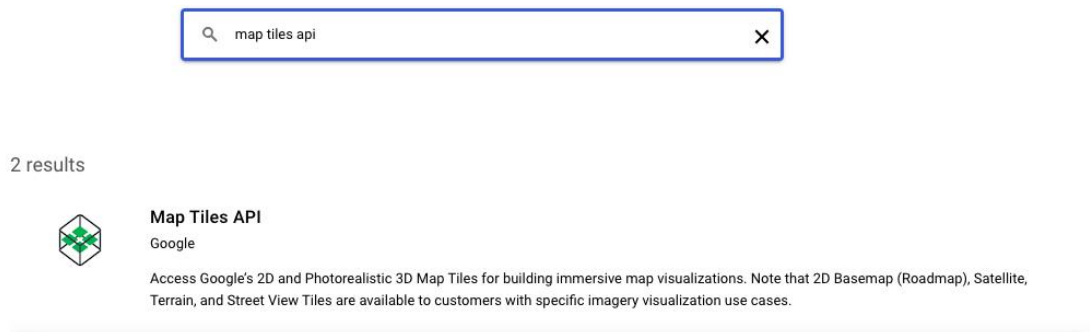
- It will appear ARCore API, select it and enable it.



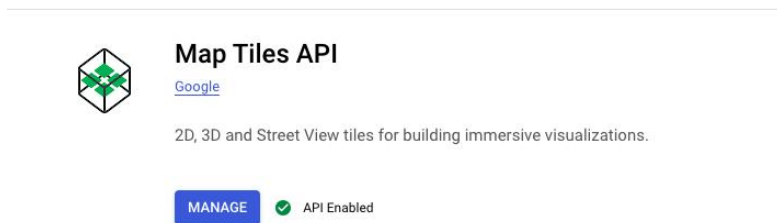
- Click manage to enable API.



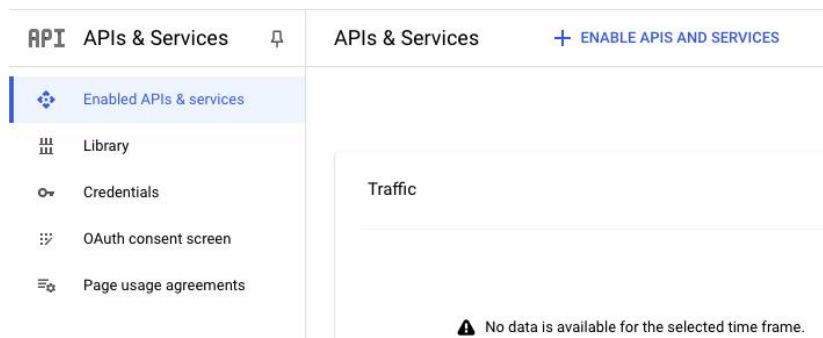
- Go to API and services page again. Search “Map tiles API” on search bar



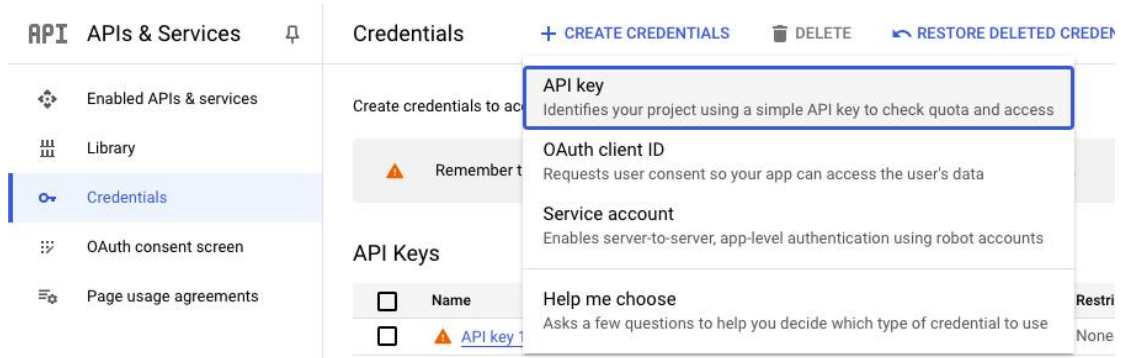
- Click it and enable the Map Tiles API. This API provides photorealistic tile view of the world inside Unity editor.



- Then we need to setup credentials. Go to API and Services page again.



- You can see Credentials tab at the left menu bar. Click it and go to Credentials page.
- Then click **Create Credentials** and select **API Key**.



- Then It will create new API key for you. Copy API key for later use.

API key created

Use this key in your application by passing it with the `key=API_KEY` parameter.

Your API key

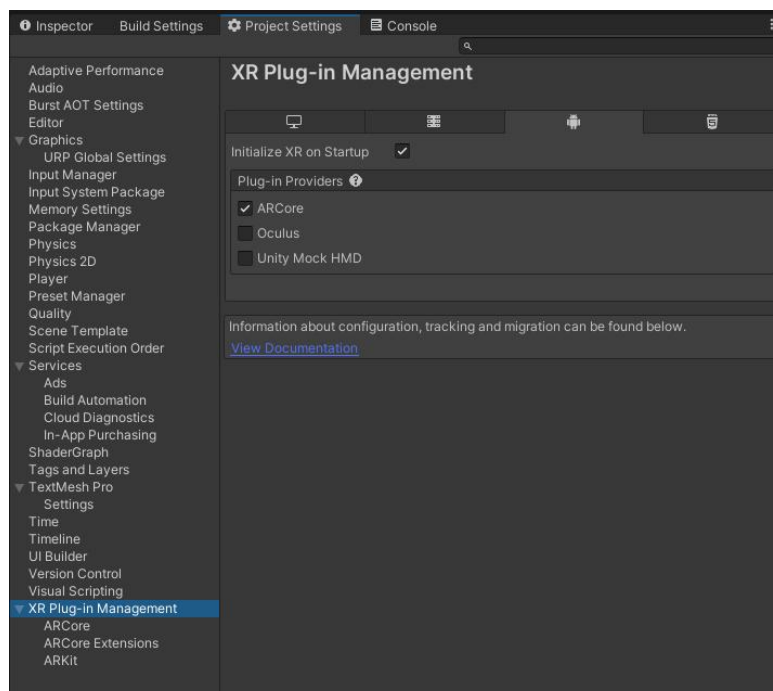
AIzaSyAPhH3goJHeShnvKW9tRQ6E3TrfVU6mAzA

[Copy to clipboard](#)

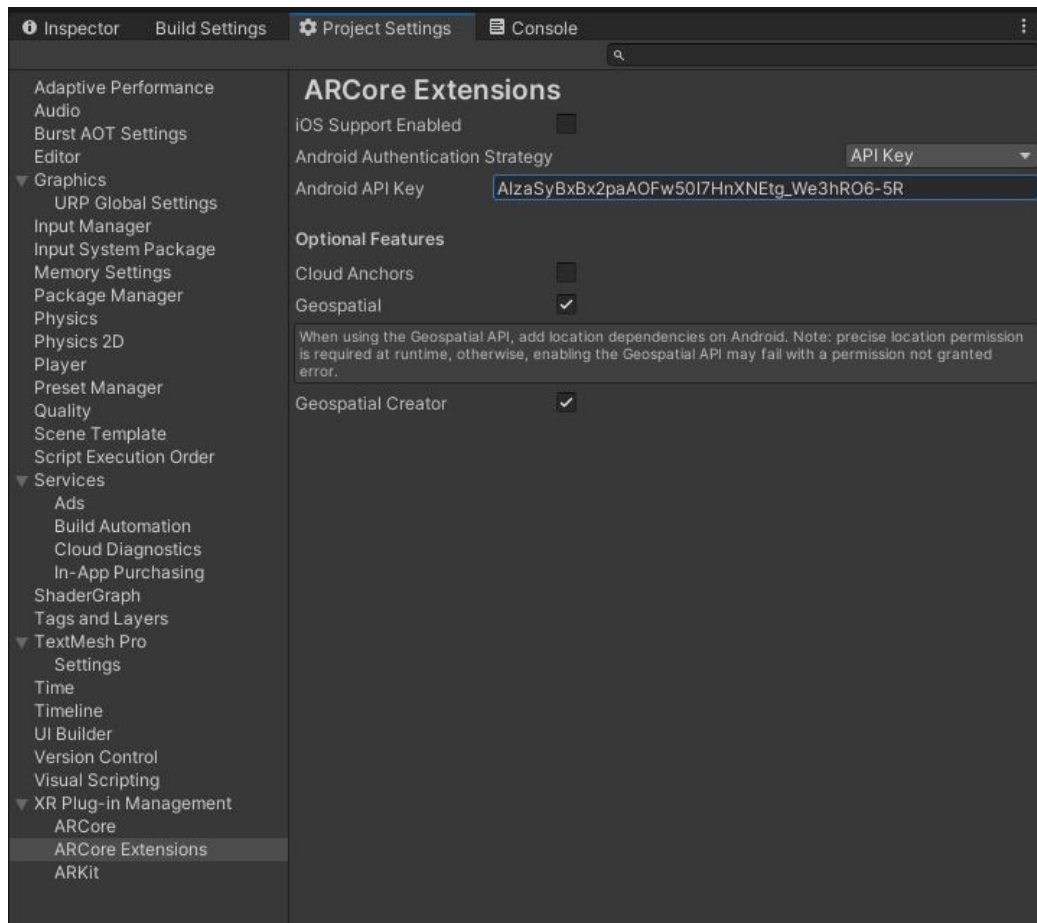
⚠ This key is unrestricted. To prevent unauthorized use, we recommend restricting where and for which APIs it can be used. [Edit API key](#) to add restrictions. [Learn more](#)

[CLOSE](#)

- In Unity project go to Player settings → XR Plug-in Management.
- Click Android tab and enable AR Core.



- Select AR Core Extensions under XR Plug-in Management. Change Android Authentication Strategy Do not use to API Key.

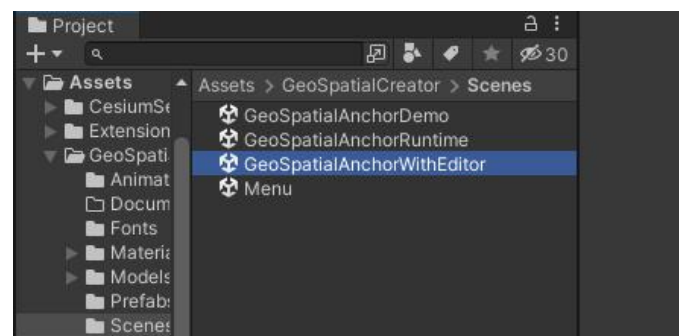


- Then Android API key text box will appear. Input the API key that you obtain from google cloud console [here](#).
- Under optional features enable Geo Spatial and Geo Spatial Creator.
- Now it's Ready to use your project with Easy AR Geo Spatial Anchors.

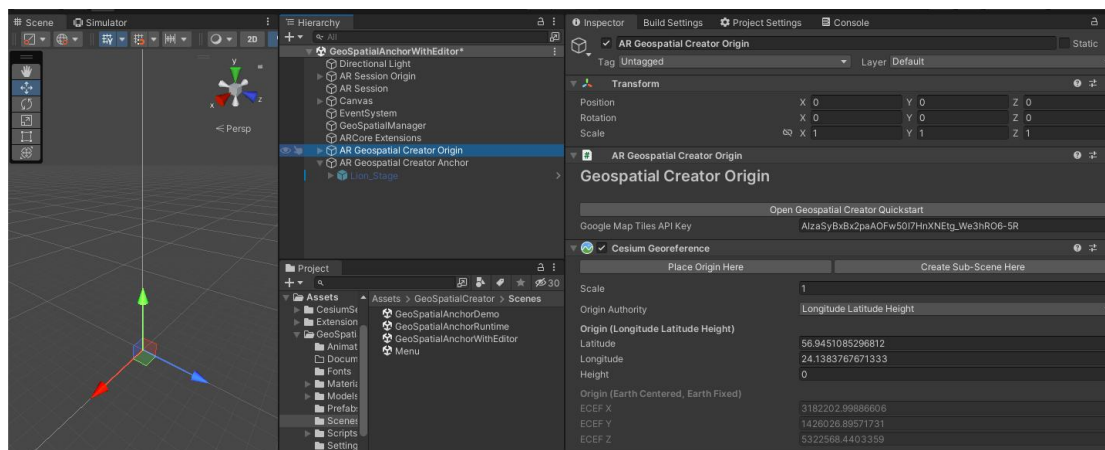
4. Add prefab on given location coordinate using Place with Editor demo

You can place 3D object in given location using GeoSpatialAnchorWithEditor demo.

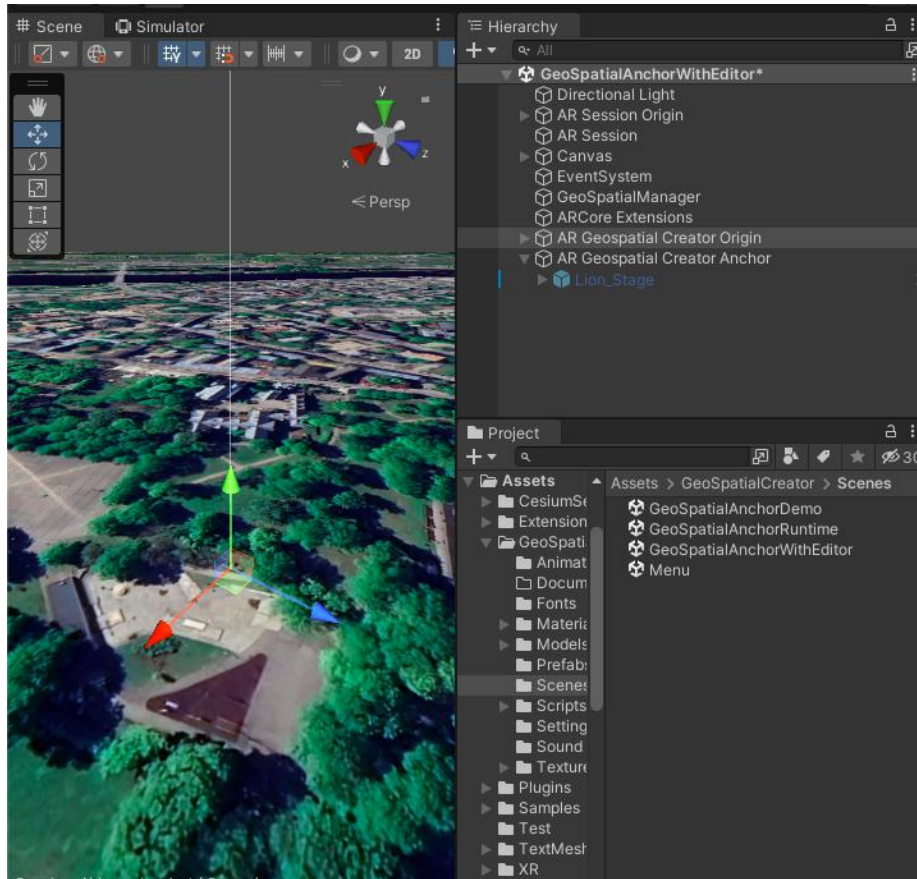
- Goto Assets → GeoSpatialCreator → Scenes folder. Double click on GeoSpatialAnchorWithEditor to load the scene.



- Once load the scene you can see heirarchy but your map tiles not loaded yet. for that select AR Geo Spatial Creator game object.



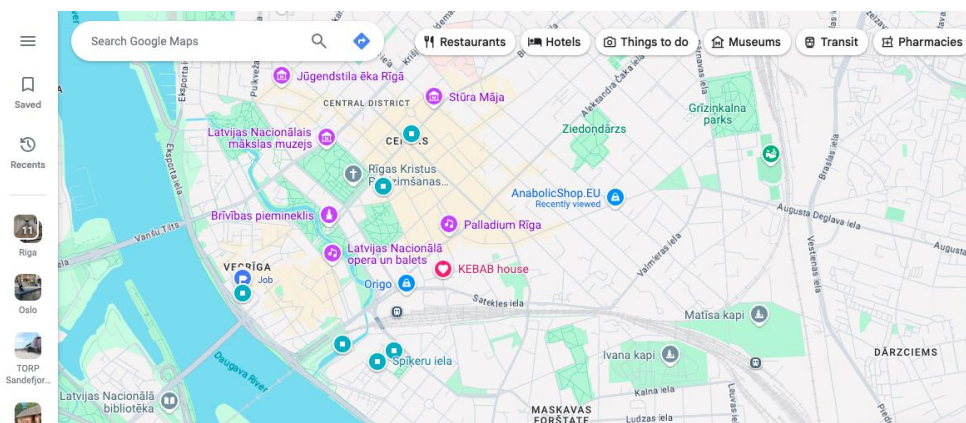
- You can see text box to input Google map tiles API Key. paste same key that you obtain from google cloud console and hit Enter. Then you will see in Scene window map tiles of world will load as follows.



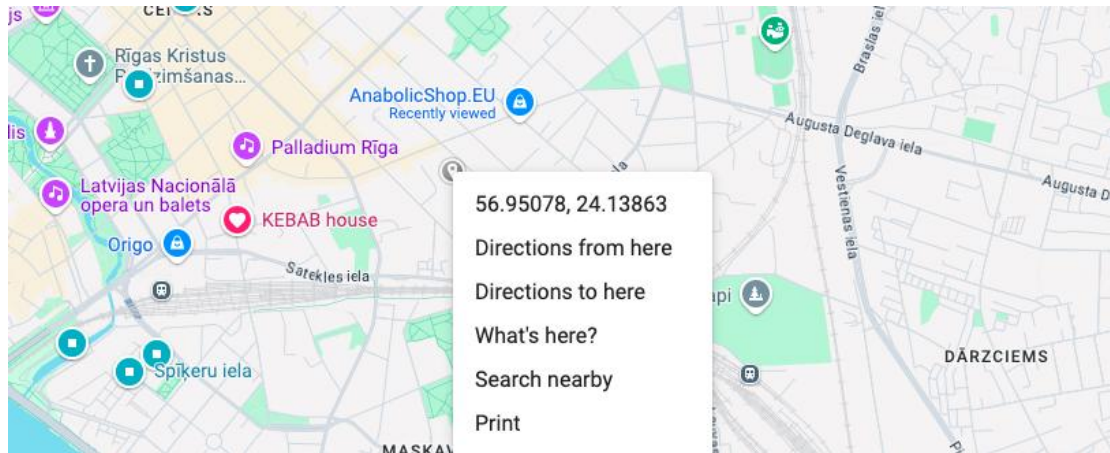
5.1. Get Location data from Google map and change location

To change the location where you want to place 3D object you have to change the AR Geo Spatial Creator game object world position. For that first you need to get location coordinates from google map.

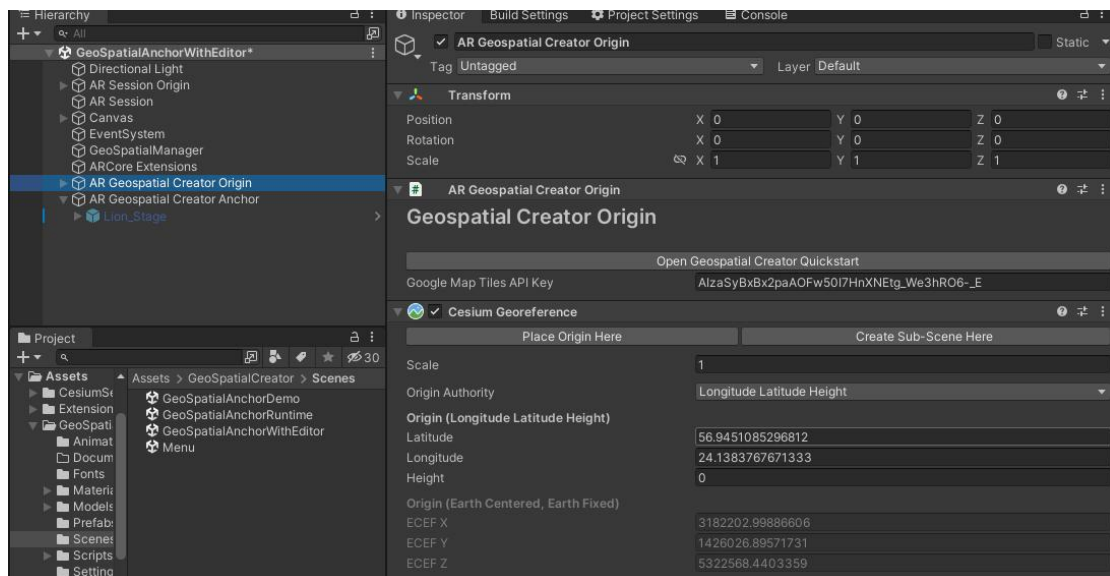
- Redirect to google maps (<https://maps.google.com/>)



- Right click where you want place your 3D object. It will appear coordinates of location at first. click on the coordinates, it will be copied to clipboard.

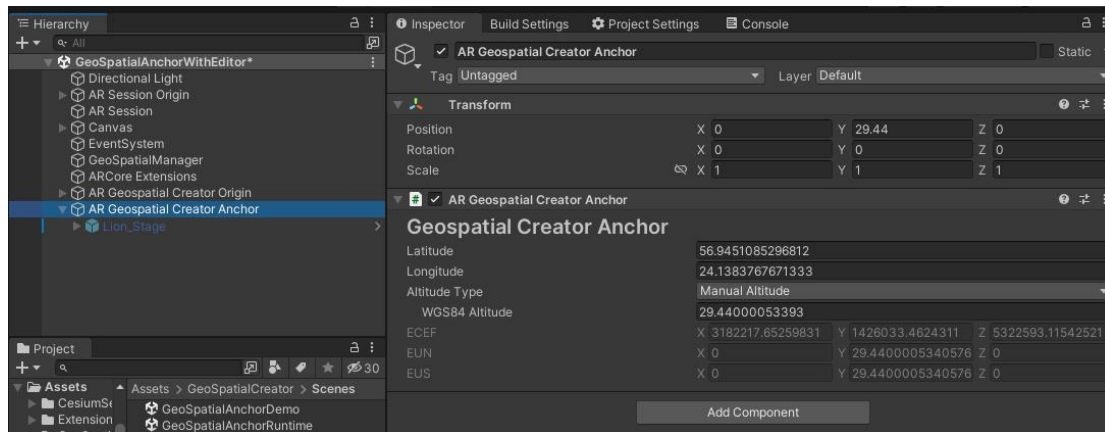


- Get back to Unity editor and select AR Geospatial Creator Origin gameobject.

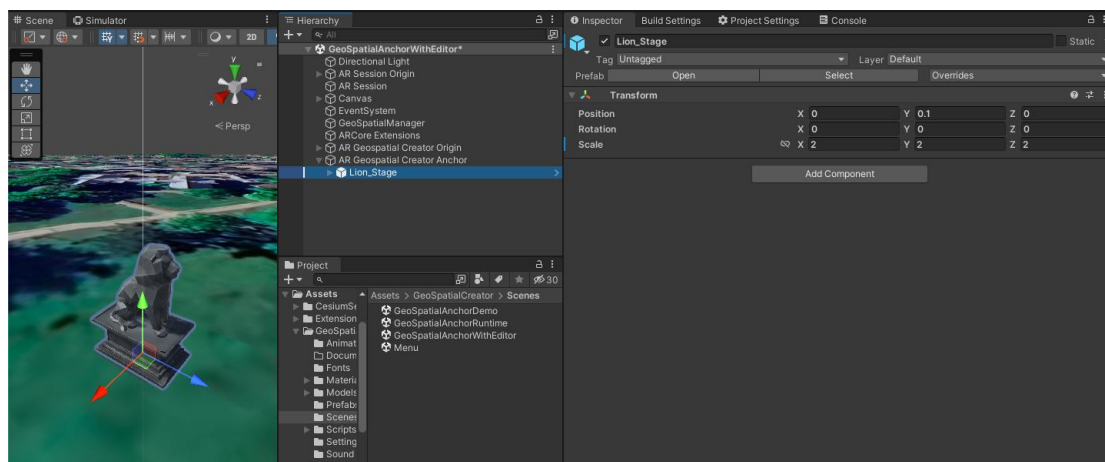


- Paste latitude and longitude values showing in Cesium Georeference component. Set height to 0.

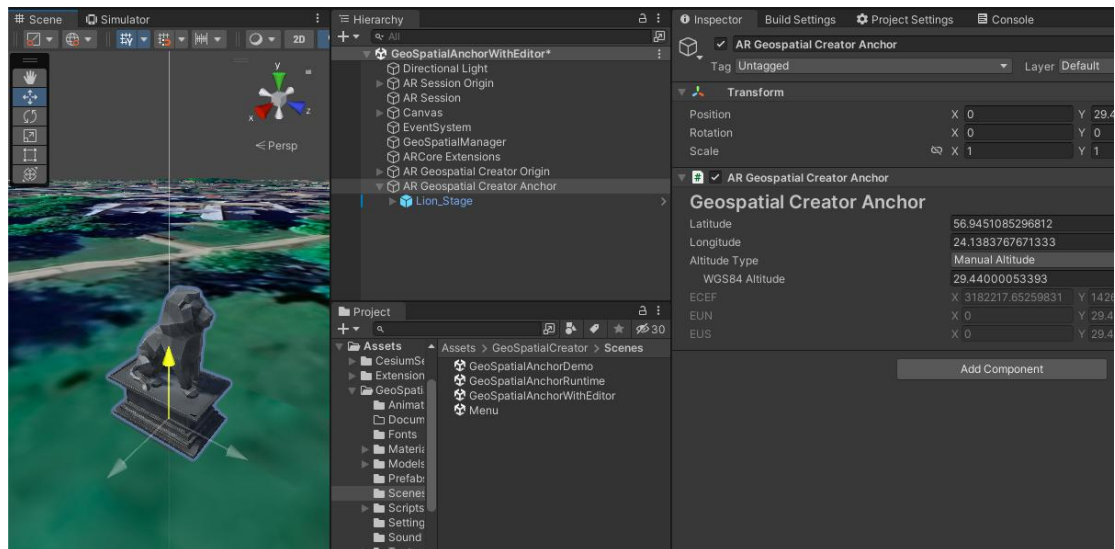
- Then Select AR GeoSPatial Creator Anchor game object. you can see Latitudes, Longitude and Altitude coordinate where current 3D object placed.



- You can adjust the location of object. Enable Lion_Stage model which child of AR GeoSPatial Creator Anchor game object.



- Double click to focus game object. If you need to adjust exact location and height of model, click AR GeoSPatial Creator Anchor game object and use move tool to change anchor location.

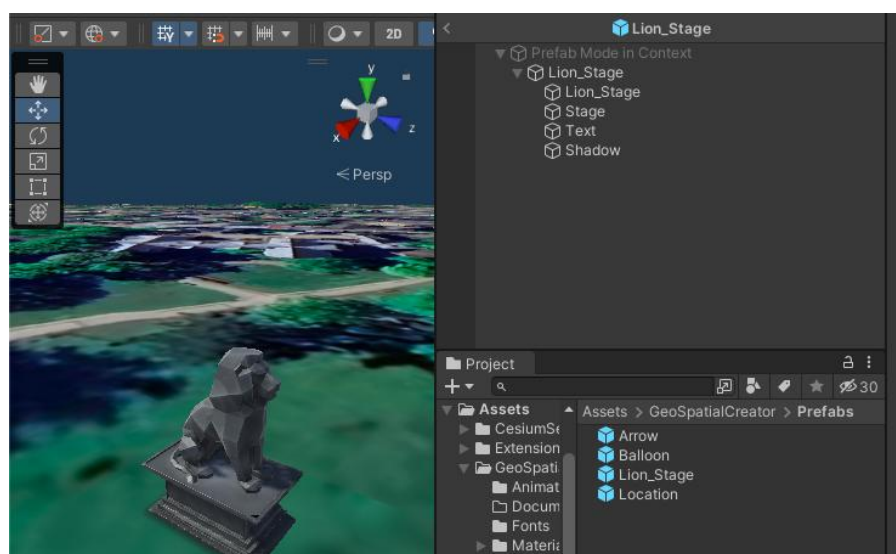


- Once you set the model right location disable the Lion_stage gameobject again.

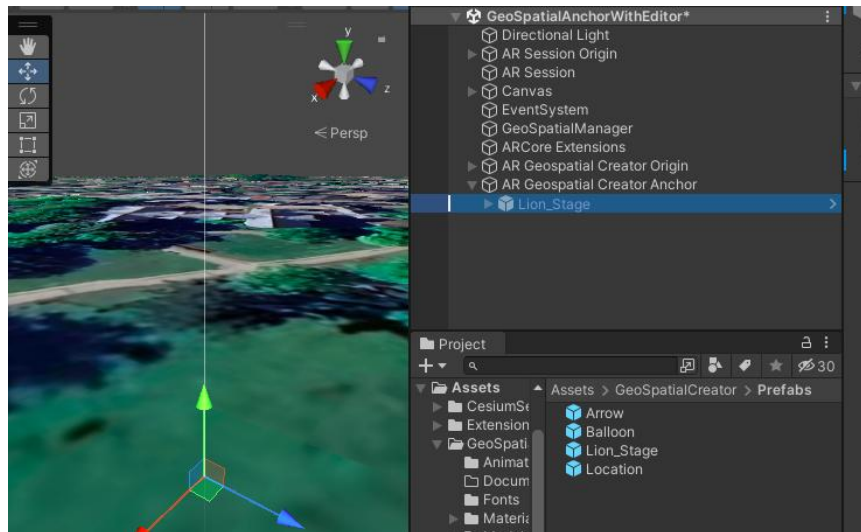
5.2. Change the 3D object

If you want to change the 3D object which you want to place it's very simple.

- By default we placed Lion_Stage game object which you can find Assets->GeoSpatialCreator->Prefabs folder.



- If you have different 3D object, you can import it into unity project
- Then scale your gameobject as you desire.
- Remove default 3D object (Lion_stage) and replace new gameobject as child of AR GeoSPatial Creator Anchor game object.



- Build the application and enjoy.

Enjoy the Easy AR Geo Spatial Anchors.. Thank you!